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Computational Entropy: Output-Distribution Entropy and Landauer-Exact Export

Paper A (solid core) — Draft Status: 2026-07-29 · Preliminary research · self-contained, no gravitational claims **Scope contract:** [../DELIVERY_SCOPE.md](#) §1 (may assert A1–A6; must not assert gravity) **Canonical definition:** [../foundations/computational-entropy/definition.md](#) **Reproducible artifacts:** `simulations/classical/m11_and_gate_ledger.py`, `m11_landauer_and_ledger.py`, `m11_composition_ledger.py`

Abstract

We define **computational entropy** as the entropy of the *output distribution* induced by a map or channel on a random input, independent of the map’s internal mechanics. Under this definition, maps that induce the same output distribution are informationally equivalent. We then give a clean thermodynamic account of the apparent local entropy reduction produced by an irreversible computation: the drop is exactly the **export** $H(X | Y)$ (the chain-rule remainder), which for a single evaluation coincides with the **Landauer** erased-bit count. We show this export bookkeeping is **path-dependent** — two circuits computing the same overall map can pay different total export — and we identify the coordinate-invariant carrier of “information imparted” as mutual information / relative entropy, with differential entropy retained only as an illustrative (coordinate-dependent) special case. All quantitative claims are backed by short, exactly-reproducing reference computations. **No gravitational or continuum claims are made in this paper.**

1. Definition

Let f be any map — deterministic, stochastic, or a quantum channel — taking an input random variable $X \sim p_X$ to an output Y . The **computational entropy** of f on p_X is the entropy of the induced marginal output distribution p_Y :

$$H_c(f; p_X) := H(Y) = - \sum_y p_Y(y) \log_2 p_Y(y) \quad (\text{discrete}),$$

$$h_c(f; p_X) := h(Y) = - \int f_Y(y) \log_2 f_Y(y) dy \quad (\text{continuous}),$$

$$S_c(\Phi; \rho_X) := S(\Phi(\rho_X)) = - \text{Tr}(\Phi(\rho_X) \log_2 \Phi(\rho_X)) \quad (\text{quantum channel}).$$

The measure depends **only** on the statistical pattern of possible outputs — not on the algorithm, source of randomness, or gate structure that produced it.

1.1 Informational equivalence (A2)

Proposition 1. *If two maps f, g induce the same output law p_Y , then $H_c(f; p_X) = H_c(g; p_X)$, and every probability functional of the output (moments, tail probabilities, quantiles) agrees.*

Example. On $U, U_1, U_2 \sim \text{Unif}[0, 1]$, both $Y_1 = \sqrt{U}$ and $Y_2 = \max(U_1, U_2)$ have density $f(y) = 2y$ on $[0, 1]$. Hence they are informationally equivalent, with

$$h_c = - \int_0^1 2y \log_2(2y) dy = -1 + \frac{1}{2 \ln 2} \approx -0.27865 \text{ bits}.$$

(The reflected map $\min(U_1, U_2)$ gives the same value by $z = 1 - y$.)

Caveat (recorded, A6). The negative value here is a feature of *differential* entropy: h is **not** invariant under smooth reparameterization of y and can be negative. Differential entropy is therefore used **illustratively only**. The invariant statements below use discrete Shannon information and, in §5, mutual information.

2. Global conservation, local transfer

A computation can lower the entropy of the visible output below the entropy of its inputs. This is not a second-law violation; it is **local transfer with global conservation**.

The AND gate. Two fair bits $X = (X_1, X_2)$, $H(X) = 2$ bits. Compute $Y = X_1 \wedge X_2$. Then

$$H(Y) = h_2(\tfrac{1}{4}) \approx 0.811278 \text{ bits}, \quad H(X | Y) \approx 1.188722 \text{ bits},$$

and the chain rule $H(X) = H(Y) + H(X | Y)$ holds exactly (reference computation: chain error $< 10^{-12}$). The visible entropy drops from 2 to 0.811 bits; the missing 1.189 bits are the **export** $H(X | Y)$ — the distinguishability between input preimages that the map discards.

3. The export identity (A3)

Theorem 1 (export = chain-rule remainder). *For any map $X \rightarrow Y$, the reduction of visible entropy from the joint input to the output is accounted for exactly by the conditional entropy of the input given the output:*

$$\underbrace{H(X)}_{\text{input}} - \underbrace{H(Y)}_{\text{output } H_c} = H(X | Y) =: \text{export}.$$

Proof. Immediate from the chain rule $H(X, Y) = H(X) = H(Y) + H(X | Y)$, using $H(X, Y) = H(X)$ because Y is a function of X (deterministic case) or by the definition of conditional entropy (stochastic case). $\square$

Define the **single-shot export/load slot** $L_S := H(X | Y)$. It is the information the environment must absorb for the computation to be realized irreversibly.

4. Landauer-exactness (A4)

Theorem 2 (Landauer coincidence). *The single-shot export $L_S = H(X | Y)$ equals the number of bits whose erasure Landauer’s principle charges: the minimum dissipated heat to realize the irreversible map at temperature T is*

$$Q \geq kT \ln 2 \cdot H(X | Y) = kT \ln 2 \cdot L_S.$$

For the AND gate, $L_S = 1.188722$ bits, so $Q \geq 1.188722 kT \ln 2$. This is not a coincidence of numerics: the erased distinguishability *is* the conditional entropy, so the export ledger and the Landauer bound are the same quantity written twice. (Reference: `m11_landauer_and_ledger.py`.)

5. Path-dependence of cumulative export (A5)

The *final* map’s export $H(X | Z)$ is path-independent (it depends only on the overall input–output relation). The **cumulative** export paid along a pipeline is **not**.

Theorem 3 (path-dependence). *There exist two pipelines computing the same overall map with equal final H_c but different total export $\sum L_S$.*

Witness. Compare a **direct** AND against a **circuit** that first publishes an intermediate wire then ANDs. Both realize the same final law ($H_{\text{final}} = 0.811278$, $H(X | Z) = 1.188722$), yet

$$\sum L_S^{\text{direct}} = 1.188722, \quad \sum L_S^{\text{circuit}} = 2.188722, \quad \text{gap} = 1.000000 \text{ bit},$$

the extra bit being the published intermediate wire $H(X | Y) = 1$. (Reference: `m11_composition_ledger.py`; composition also verifies $H(X | Y) + H(Y | Z) = H(X | Z)$ exactly and the data-processing inequality.) The active-work slot is likewise extensive ($\sum L_E$: direct 1, circuit 2), and a soft-recoverability slot L_B is non-additive across independent regions.

Interpretation. Irreversibility is charged where information is *made public / overwritten*, not merely by the abstract final function. Refactoring a computation to expose intermediates costs export even when the answer is unchanged — a discrete, exact analogue of “the way you compute matters, not just what you compute.”

6. The invariant carrier of “information imparted” (A6)

The motivating slogan — a computation makes outputs *less random*, hence *imparts information* — is made precise not by differential entropy (§1.1 caveat) but by **mutual information / relative entropy**, which are invariant under invertible reparameterization:

$$I(X; Y) = H(Y) - H(Y | X) = D_{\text{KL}}(p_{XY} \| p_X p_Y).$$

For a deterministic f , $I(X; Y) = H(Y) = H_c$; the output entropy *is* the imparted information. For stochastic maps the two separate, and $I(X; Y)$ is the correct invariant quantity. Paper A therefore

states its physical (Landauer) and structural (path-dependence) results in terms of $H(\cdot \mid \cdot)$ and $I(\cdot; \cdot)$, and treats differential entropy as a non-invariant illustration only.

7. What this paper does not claim

Per [../DELIVERY_SCOPE.md](#) §1 : no gravitational channel, load-as-time-dilation, master equation, or Gravity-from-Entropy content; no identification of toy H_c with the von Neumann entropy of any physical channel; no claim that differential entropy is itself invariant; no derivation of thermodynamics beyond the Landauer bound it instantiates.

Appendix R — Reproducibility

All numeric claims reproduce exactly (rational arithmetic on finite models):

```
.venv/bin/python simulations/classical/m11_and_gate_ledger.py      #  $H(Y)=0.811278$ ,  $export=1.18$ 
.venv/bin/python simulations/classical/m11_landauer_and_ledger.py #  $L_S = H(X/Y) = \text{Landauer error}$ 
.venv/bin/python simulations/classical/m11_composition_ledger.py  #  $path-dep: \Sigma L_S 1.188722$  vs
```

Quantity	Value	Source
$H(X)$	2 bits	AND ledger
$H_c = H(Y)$	0.811278 bits	AND ledger
$export\ H(X \mid Y)$	1.188722 bits	AND ledger
chain error	$< 10^{-12}$	AND ledger
$\sum L_S$ direct / circuit	1.188722 / 2.188722	composition ledger
path-dependence gap	1.000000 bit	composition ledger
differential h_c of $2y$	−0.27865 bits	§1.1 (illustrative)

Changelog

Date	Entry
2026-07-29	Initial draft: Def + Prop 1, Thm 1 (export), Thm 2 (Landauer), Thm 3 (path-dep), invariant framing, reproducibility.
